Expanding Brackets - Mark scheme

Algebra · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10

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Question 1 *Practice question written for this worksheet (not from a past paper)*

Expand $4(x + 5)$

$4 \times x + 4 \times 5$

Answer: 4x + 20

Question 2 *Practice question written for this worksheet (not from a past paper)*

Expand $3(2y - 7)$

$3 \times 2y - 3 \times 7$

Answer: 6y - 21

Question 3 *Practice question written for this worksheet (not from a past paper)*

Expand $x(x + 8)$

$x \times x + x \times 8$

Answer: $x^2 + 8x$

Question 4 *Practice question written for this worksheet (not from a past paper)*

Expand and simplify $3(x + 2) + 2(x - 5)$

$3x + 6 + 2x - 10$

Answer: 5x - 4

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Question 5 *Practice question written for this worksheet (not from a past paper)*

Expand and simplify $5(2x-1)-3(x+4)$

$10x-5-3x-12$

Answer: 7x - 17

Question 6 *Practice question written for this worksheet (not from a past paper)*

Expand and simplify $(x+3)(x+7)$

$x^2+7x+3x+21$

Answer: $x^2 + 10x + 21$

Question 7 *Practice question written for this worksheet (not from a past paper)*

Expand and simplify $(x-4)(x+9)$

$x^2+9x-4x-36$

Answer: $x^2 + 5x - 36$

Question 8 *Practice question written for this worksheet (not from a past paper)*

Expand and simplify $(x-5)(x-2)$

$x^2-2x-5x+10$

Answer: $x^2 - 7x + 10$

Question 9 *Practice question written for this worksheet (not from a past paper)*

Expand and simplify $(x+6)^2$

$(x+6)(x+6)=x^2+6x+6x+36$

Answer: $x^2 + 12x + 36$

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Question 10 *Practice question written for this worksheet (not from a past paper)*

Expand and simplify $(2x-3)^2$

$(2x-3)(2x-3) = 4x^2 - 6x - 6x + 9$

Answer: $4x^2 - 12x + 9$

Question 11 *Practice question written for this worksheet (not from a past paper)*

Expand and simplify $(x+5)(x-5)$

$x^2 - 5x + 5x - 25$

Answer: $x^2 - 25$

Question 12 *Practice question written for this worksheet (not from a past paper)*

Expand and simplify $(3x+2)(2x-5)$

$6x^2 - 15x + 4x - 10$

Answer: $6x^2 - 11x - 10$

Question 13 *Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2017 · Question 9*

9	$4x^2 + 6x + 6x + 9$ **or** $4x^2 + 12x + 9$		3	M1	for at least 3 terms correct in expansion of first pair of brackets
	$2x^2 - 10x + 3x - 15$ **or** $2x^2 - 7x - 15$			M1	for at least 3 terms correct in expansion of second pair of brackets **or** all 4 terms correct ignoring signs allow $-2x^2 - 7x - 15$
		$2x^2 + 19x + 24$		A1	
	Alternative method				
	$(2x+3)[(2x+3) - (x-5)]$			M1	
	$(2x+3)(x+8)$			M1	
		$2x^2 + 19x + 24$		A1	
					Total 3 marks

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Question 14

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2025 · Question 12

Q	Working	Answer	Mark	Notes		
12	$3x(2x+5)=6x^2+15x$ or $3x(7x-4)=21x^2-12x$ or $(2x+5)(7x-4)=14x^2-8x+35x-20$ $(14x^2+27x-20)$		3	M1	An expansion with only one error. Do not award this mark for $6x+15x+21x-12x$ or $(6x+15x)(21x-12x)$	M2 for 3 (out of a maximum of 4) of $42x^3+105x^2-24x^2-60x$
	$(6x^2+15x)(7x-4)=42x^3-24x^2+105x^2-60x$ $(21x^2-12x)(2x+5)=42x^3+105x^2-24x^2-60x$ $3x(14x^2-8x+35x-20)=42x^3-24x^2+105x^2-60x$ $3x(14x^2+27x-20)=42x^3+81x^2-60x$			M1	ft dep on M1 allow one further error	(M1 for 2 correct out of a maximum of 4)
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	$42x^3+81x^2-60x$		A1	cao (terms may be in any order but must be simplified) dep on M1 ISW correct factorisation eg $3(14x^3+27x^2-20x)$ Do not ISW incorrect simplification eg $14x^3+27x^2-20x$	
						Total 3 marks

Question 15

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2023 · Question 13

Q	Working	Answer	Mark	Notes
	$3x(2x-5)=6x^2-15x$ **or**		3	M1 for multiplying $3x$ by $(2x-5)$ with **both** terms correct or
	$(2x-5)^2=4x^2-10x-10x+25$ **or**			for multiplying $(2x-5)$ by $(2x-5)$ with 3 out of 4 terms correct or
	$(2x-5)^2=4x^2-20x+25$			for multiplying $(2x-5)$ by $(2x-5)$ and getting $4x^2-20x\ldots$ **or** $\ldots-20x+25$ (not for $4x^2+25$)
	$(6x^2-15x)(2x-5)=$ $12x^3-30x^2-30x^2+75x$ oe **or**			M1ft (dep) for multiplying the product of $3x$ and $(2x-5)$ by $(2x-5)$ with 3 out of 4 terms correct or
	$(6x^2-15x)(2x-5)=$ $12x^3-60x^2+75x$ oe **or**			for multiplying the product of $3x$ and $(2x-5)$ by $(2x-5)$ and getting $12x^3-60x^2\ldots$ **or** $\ldots-60x^2+75x$
	$3x(4x^2-10x-10x+25)=$ $12x^3-30x^2-30x^2+75x$ oe **or**			for multiplying the product of $(2x-5)$ and $(2x-5)$ by $3x$ with 3 out of 4 terms correct or
	$3x(4x^2-20x+25)=$ $12x^3-60x^2+75x$			for multiplying the product of $(2x-5)$ and $(2x-5)$ by $3x$ with 2 out of 3 terms correct or
				Expansion in one stage will lead to $12x^3-30x^2-30x^2+75x$ without firstly expanding two factors – award M2 for 3 out of 4 terms correct M1 for 2 out of 4 terms correct
	Working required	$12x^3-60x^2+75x$		A1 dep on M1
				Total 3 marks

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Question 16 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2021 · Question 13*

13		$4x(3x+1)=12x^2+4x$ **or** $4x(2x-3)=8x^2-12x$ **or** $(3x+1)(2x-3)=6x^2-9x+2x-3\ (=6x^2-7x-3)$		3	M1	for expanding two of the three factors, allow one error
		$(12x^2+4x)(2x-3)=24x^3-36x^2+8x^2-12x$ oe $(8x^2-12x)(3x+1)=24x^3+8x^2-36x^2-12x$ oe $4x(6x^2-7x-3)=$ eg $24x^3-28x^2....$ oe			M1	(dep)ft for expanding by the third factor, allow one error (some may do the expansion in one stage and will get to $24x^3-36x^2+8x^2-12x$ without firstly expanding two factors)
		Working required	$24x^3-28x^2-12x$		A1	dep on M1 isw correct factorisation eg $4(6x^3-7x^2-3x)$ $x(24x^2-28x-12)$ $4x(6x^2-7x-3)$ do not isw incorrect simplification eg $24x^3-28x^2-12x=6x^3-7x^2-3x$ gets M2A0
						Total 3 marks

Question 17 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · November 2020 · Question 14*

14	$(4x+1)(x-3)=4x^2-12x+x-3(=4x^2-11x-3)$ $(4x+1)(5x+6)=20x^2+24x+5x+6(=20x^2+29x+6)$ $(x-3)(5x+6)=5x^2+6x-15x-18(=5x^2-9x-18)$		3	M1	for multiplying 2 brackets with at least 3 out of 4 terms correct
	$(5x+6)(4x^2-11x-3)=20x^3-55x^2-15x+24x^2-66x-18$ $(x-3)(20x^2+29x+6)=20x^3+29x^2+6x-60x^2-87x-18$ $(4x+1)(5x^2-9x-18)=20x^3-36x^2-72x+5x^2-9x-18$			M1	(dep) for multiplying the product of the first 2 brackets (ft from the 1st stage) by the 3rd bracket, and getting at least 3 out of 6 or 4 out of 8 terms correct
		$20x^3-31x^2-81x-18$		A1	
	Alternative				
	$20x^3+24x^2-60x^2+5x^2-15x+6x-72x-18$			B2	for at least 6 out of 8 terms correct (B1 for 4 or 5 out of 8 correct terms)
		$20x^3-31x^2-81x-18$		A1	
					Total 3 marks

Question 18 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2024 · Question 15*

15	$3x(2x-1)=6x^2-3x$ or $3x(5x+4)=15x^2+12x$ or $(2x-1)(5x+4)=10x^2+8x-5x-4$ $(10x^2+3x-4)$		3	M1	An expansion with only one error. Do not award this mark for $6x-3x+15x+12x$	M2 for 3 (out of a maximum of 4) of $30x^3+24x^2-15x^2-12x$
	$(6x^2-3x)(5x+4)=30x^3+24x^2-15x^2-12x$ $(15x^2+12x)(2x-1)=30x^3-15x^2+24x^2-12x$ $3x(10x^2+8x-5x-4)=30x^3+24x^2-15x^2-12x$ $3x(10x^2+3x-4)=30x^3+9x^2-12x$			M1	ft dep on M1 allow one further error	(M1 for 2 correct out of a maximum of 4)
	Correct answer scores full marks (unless from obvious incorrect working)	$30x^3+9x^2-12x$		A1	cao (terms may be in any order but must be simplified) dep on M1 accept $a=30$, $b=9$, $c=-12$ ISW correct factorisation eg $3(10x^3+3x^2-4x)$ Do not ISW incorrect simplification	
						Total 3 marks

CHALLENGE QUESTIONS

Question 19 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2023 · Question 16*

Question	Working	Answer	Mark	Notes
16	$(2x+3)(x-5)=2x^2-10x+3x-15(=2x^2-7x-15)$ or $(2x+3)(x+4)=2x^2+8x+3x+12(=2x^2+11x+12)$ or $(x-5)(x+4)=x^2+4x-5x-20(=x^2-x-20)$ oe		3	M1 For a correct method to expand two brackets with at least 3 terms correct out of 4 terms (or 2 terms correct out of 3 terms) Do not award this mark for eg $2x^2-10x+3x-15+x^2+4x-5x-20$ or eg $2x^2-10x+3x-15+x+4$
	$(2x^2-7x-15)(x+4)=2x^3+8x^2-7x^2-28x-15x-60$ or $(2x^2+11x+12)(x-5)=2x^3-10x^2+11x^2-55x+12x-60$ or $(x^2-x-20)(2x+3)=2x^3+3x^2-2x^2-3x-40x-60$ oe			M1ft Ft dep on M1 and a quadratic for a correct method to multiply by the 3rd bracket – allow one further error
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	$2x^3+x^2-43x-60$		A1 If no working shown then award B2 for 3 out of a maximum of 4 terms correct
	ALTERNATIVE			
	$2x^3+8x^2-10x^2-40x+3x^2+12x-15x-60$		3	M2 For a complete expansion with 8 terms present of which 4 are correct (M1 for 4 correct terms from any number of terms)
		$2x^3+x^2-43x-60$		A1
				Total 3 marks

Question 20 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2022 · Question 13*

Q	Working	Answer	Mark	Notes
(a)	$5x(x+2)=5x^2+10x$ **or** $(x+2)(3x-4)=$ $3x^2-4x+6x-8$ $(=3x^2+2x-8)$ **or** $5x(3x-4)=15x^2-20x$		3	M1 for a correct intention to multiply all 3 factors by starting to multiply 2 factors only, allow one error
	eg $[(5x^2+10x)(3x-4)=]$ $15x^3-20x^2+30x^2-40x$ **or** $[5x(3x^2+2x-8)=]$ $15x^3+10x^2-40x$ **or** $[(x+2)(15x^2-20x)=]$ $15x^3-20x^2+30x^2-40x$			M1 (dep)ft for expanding by the third factor, allow one error (some may do the expansion in one stage and will get to $15x^3-20x^2+30x^2-40x$ without firstly expanding two factors, allow two errors)
		$15x^3+10x^2-40x$		A1 isw correct factorisation eg $5(3x^3+2x^2-8x)$ do not isw incorrect factorisation eg $15x^3+10x^2-40x=3x^3+2x^2-8x$

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Q	Working	Answer	Mark	Notes
(b)	$\left(\frac{2w^2}{y^5}\right)^{-3}$ or $\left(\frac{y^{20}}{16w^8}\right)^{\frac{3}{4}}$ or $\left(\frac{4096w^{24}}{y^{60}}\right)^{-\frac{1}{4}}$		3	M1 for one of fourth rooting or reciprocating or cubing
	$\left(\frac{8w^6}{y^{15}}\right)^{-1}$ or $\frac{2^{-3}w^{-6}}{y^{-15}}$ or $\frac{\frac{1}{8}w^{-6}}{y^{-15}}$ or $\left(\frac{y^5}{2w^2}\right)^3$ or $\left(\frac{y^{60}}{4096w^{24}}\right)^{\frac{1}{4}}$ or $\frac{0.125y^{15}}{w^6}$ or $\frac{0.125w^{-6}}{y^{-15}}$ or $\frac{0.125}{y^{-15}w^6}$ oe			M1 for two of fourth rooting or reciprocating or cubing
		$\frac{y^{15}}{8w^6}$		A1 allow $\frac{y^{15}}{8w^6}$ or $\frac{y^{15}w^{-6}}{8}$ or $0.125y^{15}w^{-6}$ or $\frac{1}{8}y^{15}w^{-6}$ or $\frac{w^{-6}}{8y^{-15}}$ or $\frac{1}{8y^{-15}w^6}$

Q	Working	Answer	Mark	Notes
	ALTERNATIVE		3	M2 for 2 correct terms (M1 for 1 correct term)
		$\frac{y^{15}}{8w^6}$		A1 allow $\frac{y^{15}}{8w^6}$ or $\frac{y^{15}w^{-6}}{8}$ or $0.125y^{15}w^{-6}$ or $\frac{1}{8}y^{15}w^{-6}$ or $\frac{w^{-6}}{8y^{-15}}$ or $\frac{1}{8y^{-15}w^6}$
				Total 6 marks

Question 21 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2025 · Question 14*

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Q	Working	Answer	Mark	Notes		
(a)	eg $7x(3x+2)=21x^2+14x$ **or** $7x(2x-5)=14x^2-35x$ **or** $(3x+2)(2x-5)=$ $6x^2-15x+4x-10$ $(=6x^2-11x-10)$		3	M1	an expansion with only one error. Do not award this mark for $21x^2+14x+14x^2-35x$ **or** $(21x^2+14x)(14x^2-35x)$	M2 for 3 (out of a maximum of 4) of $42x^3-105x^2+28x^2-70x$
	eg $42x^3-105x^2+28x^2-70x$ **or** $42x^3-77x^2-70x$			M1	ft dep on M1 and a quadratic expression allow one further error	(M1 for 2 correct out of a maximum of 4)
	Correct answer scores full marks (unless from obvious incorrect working)	$42x^3-77x^2-70x$		A1	if no working shown then award B2 for 2 terms out of a maximum of 3 terms correct isw correct factorisation eg $7(6x^3-11x^2-10x)$ provided 3 marks has been awarded do not isw incorrect simplification eg $42x^3-77x^2-70x=6x^3-11x^2-10x$ gets M2A0	
(b)	eg $\frac{9\times7}{7\times2y}+\frac{5\times2y}{7\times2y}(=5)$ oe or $\frac{63}{14y}+\frac{10y}{14y}(=5)$		3	M1	for writing LHS correctly over the same common denominator **or** for subtracting $\frac{5}{7}$ from both sides allow use of equivalent	

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	oe **or** $\frac{63+10y}{14y}(=5)$ oe **or** $\frac{9}{2y}=5-\frac{5}{7}$ **or** $\frac{9}{2y}=4\frac{2}{7}\left(=\frac{30}{7}\right)$ **or** $9+\frac{10y}{7}=10y$ **or** $\frac{63}{2y}+5=35$				decimal 0.71(42…) for method marks **or** multiplying through by $2y$ **or** multiplying through by 7
	eg $63+10y=70y$ oe **or** $63=60y$ oe **or** $\frac{63}{2}=30y$ **or** $2.1=2y$ **or** $2y=\frac{9}{30/7}$ oe			M1	for a correct equation with all fractions removed **or** for a correct equation with y isolated
	Working required	1.05		A1	oe eg $\frac{63}{60}$ or $\frac{21}{20}$ dep on M1
					Total 6 marks

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Question 22

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2024 · Question 14

14 (a)	$(3x+1)(2-x) = 6x-3x^2+2-x\ (=-3x+5x+2)$ **or** $(2-x)(4+x) = 8+2x-4x-x\ \ (=-x^2-2x+8)$ **or** $(3x+1)(4+x) = 12x+3x+4+x\ (=3x+13x+4)$		3	M1	for a correct method to expand two brackets with at least 3 terms correct out of 4 terms (or 2 terms correct out of 3 terms). Do not award this mark for eg $6x-3x^2+2-x+8+2x-4x-x$ or $6x-3x^2+2-x+4+x$
	$(-3x+5x+2)(4+x) = -12x^2-3x^3+20x+5x+8+2x$ **or** $(-x^2-2x+8)(3x+1) = -3x^3-x^2-6x-2x+24x+8$ **or** $(3x+13x+4)(2-x) = 6x^2-3x+26x-13x^2+8-4x$			M1	ft dep on M1 **and** a quadratic for a correct method to multiply by the 3rd bracket – allow one further error
	Correct answer scores full marks (unless from obvious incorrect working)	$-3x^3-7x^2+22x+8$		A1	oe but must be simplified eg $22x-3x^3-7x^2+8$ if no working shown then award B2 for 3 out of a maximum of 4 terms correct
	ALTERNATIVE				
ALT (a)	$24x+6x^2-12x^2-3x^3+8+2x-4x-x^2$ oe		3	M2 (M1	for a complete expansion with 8 terms present of which at least 4 are correct for at least 4 correct terms from any number of terms)
		$-3x^3-7x^2+22x+8$		A1	oe but must be simplified eg $22x-3x^3-7x^2+8$ if no working shown then award B2 for 3 out of a maximum of 4 terms correct

(b)	$\left(\frac{1}{a^6b^4}\right)^{-\frac{1}{2}}$ **or** $\left(a^{-6}b^{-4}\right)^{-\frac{1}{2}}$ **or** $\left(\frac{a^{1.5}b^{0.5}}{a^{4.5}b^{2.5}}\right)^{-1}$ **or** $\left(\frac{a^9b^5}{a^3b}\right)^{\frac{1}{2}}$ **or** $\left(\frac{a^{-3}b^{-1}}{a^{-9}b^{-5}}\right)^{\frac{1}{2}}$ oe		3	M1	for simplifying the a and the b term in the fraction **or** for applying the power ½ to at least 3 out of 4 of a^3, b, a^9, b^5 **or** for applying the negative power to at least 3 out of 4 of a^3, b, a^9, b^5
	$\left(\frac{1}{a^3b^2}\right)^{-1}$ **or** $\left(a^{-3}b^{-2}\right)^{-1}$ **or** $\frac{a^{4.5}b^{2.5}}{a^{1.5}b^{0.5}}$ **or** $\left(a^6b^4\right)^{\frac{1}{2}}$ **or** $\left(\frac{a^{-1.5}b^{-0.5}}{a^{-4.5}b^{-2.5}}\right)$ oe			M1	for two of simplifying the a and the b term in the fraction **or** for applying the power ½ to at least 3 out of 4 of a^3, b, a^9, b^5 **or** for applying the negative power to at least 3 out of 4 of a^3, b, a^9, b^5
	Correct answer scores full marks (unless from obvious incorrect working)	a^3b^2		A1	accept $\frac{1}{a^{-3}b^{-2}}$
					Total 6 marks

Question 23

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2022 · Question 16

16	(a)	$(3x-1)(x+2)=3x^2+6x-x-2(=3x^2+5x-2)$ **or** $(3x-1)(3x+1)=9x^2+3x-3x-1(=9x^2-1)$ **or** $(x+2)(3x+1)=3x^2+x+6x+2(=3x^2+7x+2)$		3	M1 for a correct intention to multiply all 3 factors by multiplying 2 factors only, allow one error
		$[(3x^2+5x-2)(3x+1)=]\ 9x^3+15x^2-6x+3x^2+5x-2$ **or** $[(9x^2-1)(x+2)=]\ 9x^3+18x^2-x-2$ **or** $[(3x^2+7x+2)(3x-1)=]\ 9x^3+21x^2+6x-3x^2-7x-2$			M1 (dep)ft for expanding by the third factor, allow one error
			$9x^3+18x^2-x-2$		A1
		ALTERNATIVE			
		$9x^3+3x^2+18x^2+6x-3x^2-x-6x-2$		3	M2 for a complete expansion with 8 terms present, at least 4 of which must be correct
			$9x^3+18x^2-x-2$		A1
	(b)	$\left(\frac{8xy^2}{2x^5}\right)^2$ **or** $\left(\frac{x^4}{4y^2}\right)^{-2}$ **or** $\left(\frac{4x^{10}}{64x^2y^4}\right)^{-1}$		3	M1 for one of reciprocating or simplifying or squaring
		$\left(\frac{4y^2}{x^4}\right)^2$ **or** $\left(\frac{x^8}{16y^4}\right)^{-1}$ **or** $\frac{64x^2y^4}{4x^{10}}$ **or** $\frac{\frac{1}{4}x^{-10}}{\frac{1}{64}x^{-2}y^{-4}}$			M1 for two of reciprocating or simplifying or squaring
			$\frac{16y^4}{x^8}$		A1 accept $16y^4x^{-8}$ or $\frac{16}{y^{-4}x^8}$ or $\frac{16x^{-8}}{y^{-4}}$ oe
		ALTERNATIVE			
				3	M2 for 2 correct terms (M1 for 1 correct term)
			$\frac{16y^4}{x^8}$		A1 accept $16y^4x^{-8}$ or $\frac{16}{y^{-4}x^8}$ or $\frac{16x^{-8}}{y^{-4}}$ oe
					Total 6 marks

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